PROJECT PROPOSAL

AI Interview Practice and Evaluation System

An AI-powered web application for interactive interview practice, automated answer evaluation, and personalized feedback through voice and text.

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Program	BS Information Technology (BS IT)
Project Type	Final Year Project (FYP)

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Prepared for academic review. Technologies and implementation details may be refined with supervisor guidance.

1. Introduction

The AI Interview Practice and Evaluation System is an AI-powered web application designed to help students, fresh graduates, and job seekers prepare for interviews in an interactive and effective way. Users will be able to respond to interview questions through either voice or text. The system will assess their responses based on relevance, clarity, completeness, and communication quality. For voice-based interviews, it may also analyze speech-related features such as speaking pace and pauses.

The main purpose of this project is to provide an accessible virtual interview environment where users can improve their interview skills, identify weaknesses, and build confidence before attending real job interviews.

2. Problem Statement

Many students and fresh graduates face difficulties during job interviews because they lack interview practice, confidence, and knowledge of how to present their answers effectively. Professional interview coaching and mock interview sessions may be expensive, time-consuming, or difficult to access.

Therefore, there is a need for an AI-powered interview practice system that allows users to conduct mock interviews, receive automated evaluations, and obtain personalized feedback in a convenient online environment.

3. Proposed Solution

The proposed system will provide a web-based mock interview environment. Users will select an interview category or role, answer generated questions using text or voice, and receive structured feedback after each response or interview session. The system will present evaluation scores and practical suggestions to help users improve future attempts.

4. Objectives

• Provide an accessible platform for practicing job interviews online.

• Generate relevant interview questions according to a selected role or topic.

• Support both text-based and voice-based responses.

• Evaluate answers using predefined criteria such as relevance, clarity, completeness, and structure.

• Provide personalized feedback and suggestions for improvement.

• Track interview history and display progress over time.

• Help users build confidence and improve interview readiness.

5. Scope of the Project

The initial version will focus on individual mock interview sessions for students, fresh graduates, and job seekers. It may include user registration and login, role-based question generation, text and voice input, automated answer evaluation, session reports, and a progress dashboard. Advanced emotion detection, facial-expression analysis, or real-time video assessment are not included unless time and resources permit.

6. Proposed Technologies

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Frontend	HTML, CSS, JavaScript, React.js
Backend	Python with FastAPI
Database	PostgreSQL or MySQL
AI question generation	Large Language Model (LLM) API
Answer evaluation	LLM-based evaluation using predefined rubrics
Speech-to-text	Whisper or a suitable speech recognition API
Text-to-speech	Browser SpeechSynthesis or a suitable TTS service
Authentication	JWT-based authentication
Data visualization	Chart.js or Recharts
Version control	Git and GitHub
Development environment	Visual Studio Code

7. Main System Modules

User Management

Registration, login, and basic profile management.

Interview Setup

Selection of job role, interview category, and difficulty level.

Question Generation

Generation of role-relevant questions using an LLM API.

Answer Capture

Text input and voice recording/transcription where supported.

Answer Evaluation

Scoring based on a consistent rubric and explanation of the score.

Feedback and Suggestions

Strengths, weaknesses, and actionable improvement tips.

Progress Dashboard

Previous sessions, score trends, and performance summaries.

8. Expected Outcomes

- A working web application for mock interview practice.
- AI-generated interview questions tailored to selected roles or topics.
- Automated evaluation reports for text and transcribed voice answers.
- Personalized feedback to help users improve answer quality.
- A dashboard for reviewing previous sessions and tracking progress.

9. Conclusion

The AI Interview Practice and Evaluation System aims to make interview preparation more accessible, interactive, and personalized. By combining web technologies, Artificial Intelligence, Natural Language Processing, and speech recognition, the proposed system will allow users to practice interviews, receive automated feedback, and monitor their progress. The project will provide practical experience in software engineering, AI integration, database design, web application development, and software testing. It is intended to serve as a useful academic Final Year Project with potential real-world applications for students and job seekers.